Data Analysis & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Mean, median, mode and range

Find the mean of $6, 9, 12, 15, 18$.

Mean $= \frac{6+9+12+15+18}{5} = \frac{60}{5} = 12$.

2 [Foundational] • Target: Mean, median, mode and range

Ages of 7 students: $13, 15, 13, 16, 14, 13, 17$. Find the mean, median, and mode.

Ordered: $13, 13, 13, 14, 15, 16, 17$. Mean $= \frac{101}{7} \approx 14.43$. Median (4th value) $= 14$. Mode $= 13$ (appears three times).

3 [Foundational] • Target: Mean, median, mode and range

Find the range and mode of the data set $8, 3, 8, 15, 8, 22, 3$.

Range $= 22 - 3 = 19$. Mode $= 8$ (appears three times).

4 [Foundational] • Target: Mean, median, mode and range

A data set has mean 18 and 6 values. Find the sum of the values.

Sum $= 18 \times 6 = 108$.

5 [Foundational] • Target: Mean, median, mode and range

The data set $10, 12, 11, 13, 50$ contains an outlier. Find the mean before and after removing 50.

Before: $\frac{10+12+11+13+50}{5} = \frac{96}{5} = 19.2$. After removing 50: $\frac{46}{4} = 11.5$.

6 [Foundational] • Target: Mean, median, mode and range

A data set of 5 values has mean 20. Four of the values are $15, 18, 22, 25$. Find the fifth value.

Total sum $= 20 \times 5 = 100$. Known sum $= 15 + 18 + 22 + 25 = 80$. Fifth value $= 100 - 80 = 20$.

7 [Foundational] • Target: Reading tables

A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?

$\frac{12}{50} = \frac{6}{25}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Reading tables**

Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?

$15 + 13 = 28$ people. $\frac{28}{50} \times 100\% = 56\%$.

9 **[Intermediate]** • **Target: Reading tables**

A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.

Mode $= 8$ (8 students, the highest frequency). Total students $= 3 + 5 + 8 + 4 + 2 = 22$.

10 **[Intermediate]** • **Target: Reading tables**

Using the same quiz data, find the mean score to two decimal places.

Sum $= 5(3) + 7(5) + 8(8) + 9(4) + 10(2) = 15 + 35 + 64 + 36 + 20 = 170$. Mean $= \frac{170}{22} \approx 7.73$.

11 **[Intermediate]** • **Target: Reading bar charts**

The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.

Total $= 45 + 30 + 20 + 15 = 110$ books. The modal genre is Fiction, with 45 books.

12 **[Intermediate]** • **Target: Reading bar charts**

Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

$\frac{30}{110} \times 100\% \approx 27\%$.

13 **[Intermediate]** • **Target: Reading bar charts**

The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.

Mean $= \frac{60 + 45 + 80 + 55}{4} = \frac{240}{4} = 60$ mm.

14 **[Intermediate]** • **Target: Reading bar charts**

Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

Decrease $= 80 - 55 = 25$ mm. $\frac{25}{80} \times 100\% = 31.25\%$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Word Problems & Reasoning

A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.

$P(\text{red}) = \frac{4}{10} = \frac{2}{5}$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Arithmetic & Number Sense

A fair six-sided die is rolled. Find $P(\text{a number greater than } 4)$.

Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.

$P = \frac{13}{52} = \frac{1}{4}$.

18 [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Algebra Essentials

A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.

Even outcomes $\{2, 4, 6, 8\}$: $P = \frac{4}{8} = \frac{1}{2}$.

19 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.

$P(\text{red}) = \frac{3}{10}$, so $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$.

20 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 5.01 Quantitative Comparisons

Two fair coins are tossed. Find $P(\text{both land heads})$.

$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Mean, median, mode and range

The bar chart shows quiz scores. Find the total number of students, and the modal score.

a. 20 **students; mode** $= 8$ ✓

b. 20 students; mode $= 6$

c. 20 students; mode $= 9$

d. 18 students; mode $= 8$

Total $= 2 + 4 + 6 + 5 + 3 = 20$; the score 8 has the highest frequency (6 students).

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22 [Foundational] • Target: Reading bar charts

The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.

a. 120; Romance

b. 100; Fiction

c. 120; Mystery

d. 120; **Fiction** ✓

Total $= 45 + 30 + 25 + 20 = 120$; Fiction had the most sales (45).

23 [Foundational] • Target: Reading bar charts

The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.

a. 55; Store B

b. 55; **Store D** ✓

c. 140; Store D

d. 35; Store D

Range $= 140 - 85 = 55$; Store D had the highest sales (140).

24 [Foundational] • Target: Reading bar charts

The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.

a. 12; 2

b. 12; 2.4 ✓

c. 12; 3

d. 10; 2

Total $= 2 + 3 + 1 + 4 + 2 = 12$; mean $= 12 \div 5 = 2.4$.

25 [Foundational] • Target: Reading bar charts

The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.

a. 40; 10 ✓

b. 40; 7

c. 35; 10

d. 40; 15

Total $= 12 + 8 + 15 + 5 = 40$; range $= 15 - 5 = 10$.

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26 [Foundational] • Target: Word problems

The bar chart shows a company's quarterly sales, in thousands of riyals. Find the total annual sales, and the mean quarterly sales.

a. 1,560; 450

b. 1,560; 390 ✓

c. 1,560; 400

d. 1,200; 390

Total $= 320 + 410 + 380 + 450 = 1{,}560$; mean $= 1{,}560 \div 4 = 390$.

27 [Foundational] • Target: Mean, median, mode and range

Ages of 6 students: $12, 14, 12, 15, 13, 12$. Find the mean, median, and mode.

a. Mean $= 13$, median $= 12$, mode $= 12.5$

b. **Mean** $= 13$, **median** $= 12.5$, **mode** $= 12$ ✓

c. Mean $= 12.5$, median $= 13$, mode $= 12$

d. Mean $= 13$, median $= 12.5$, mode $= 13$

Sum $= 78$, mean $= 13$. Sorted: $12, 12, 12, 13, 14, 15$; median $= \frac{12+13}{2} = 12.5$; mode $= 12$ (appears 3 times).

28 [Foundational] • Target: Mean, median, mode and range

A data set of 6 values has mean 25. Five of the values are $20, 22, 28, 30, 32$. Find the sixth value.

a. 18 ✓

b. 28

c. 8

d. 150

Total needed $= 6 \times 25 = 150$; sum of five $= 132$; sixth $= 150 - 132 = 18$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Reading tables

A survey of 60 people's favorite drink: Tea 18, Coffee 22, Juice 12, Water 8. What fraction of people chose coffee?

a. $\frac{18}{60}$

b. $\frac{22}{60}$

c. $\frac{11}{30}$ ✓

d. $\frac{11}{60}$

$\frac{22}{60}$ simplifies (divide by 2) to $\frac{11}{30}$.

30 [Intermediate] • Target: Reading tables

Using the same survey (Tea 18, Coffee 22, Juice 12, Water 8, out of 60), what percent chose either tea or water, to the nearest percent?

a. 43% ✓

b. 30%

c. 50%

d. 26%

$(18 + 8) \div 60 = 26 \div 60 \approx 43\%$.

31 [Intermediate] • Target: Reading tables

A table shows test scores: score 60 (4 students), 70 (7), 80 (9), 90 (6), 100 (4). Find the total number of students.

a. 34

b. 30 ✓

c. 26

d. 25

$4 + 7 + 9 + 6 + 4 = 30$.

32 [Intermediate] • Target: Reading tables

Using the same test data, find the mean score to two decimal places.

a. 79.67 ✓

b. 80.00

c. 76.33

d. 79.00

Sum $= 240 + 490 + 720 + 540 + 400 = 2{,}390$; mean $= 2{,}390 \div 30 \approx 79.67$.

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33 [Intermediate] • Target: Reading tables

A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.

a. 0

b. 1.2

c. 2

d. 1 ✓

With 25 values sorted, the median is the 13th value, which falls in the '1 pet' group.

34 [Intermediate] • Target: Reading tables

Using the same pet data, find the mean number of pets per family, to two decimal places.

a. 1.00

b. 1.20 ✓

c. 1.50

d. 30

Sum $= 0(6) + 1(10) + 2(7) + 3(2) = 30$; mean $= 30 \div 25 = 1.20$.

35 [Intermediate] • Target: Reading bar charts

Using the same rainfall data (Jan 50, Feb 70, Mar 40, Apr 65 mm), find the percent decrease from Feb to Mar, to the nearest percent.

a. 75%

b. 43% ✓

c. 30%

d. 40%

Decrease $= 70 - 40 = 30$; $30 \div 70 \approx 43\%$.

36 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{a number less than } 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{3}$ ✓

Favorable outcomes $\{1, 2\}$: $\frac{2}{6} = \frac{1}{3}$.

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37 [Intermediate] • Target: Basic probability

A card is drawn from a standard deck of 52 cards. Find $P(\text{a king})$.

a. $\frac{1}{4}$

b. $\frac{1}{52}$

c. $\frac{1}{13}$ ✓

d. $\frac{4}{13}$

4 kings out of 52: $\frac{4}{52} = \frac{1}{13}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Algebra Essentials

A spinner has 10 equal sections numbered 1 to 10. Find $P(\text{a multiple of 3})$.

a. $\frac{1}{10}$

b. $\frac{1}{3}$

c. $\frac{4}{10}$

d. $\frac{3}{10}$ ✓

Multiples of 3 from 1–10: $\{3, 6, 9\}$, so $\frac{3}{10}$.

39 [Challenging] • Target: Basic probability • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 6 yellow and 9 green balls. Find $P(\text{yellow})$.

a. $\frac{1}{5}$

b. $\frac{6}{9}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$ ✓

6 out of 15 simplifies to $\frac{2}{5}$.

40 [Challenging] • Target: Basic probability • Cross-Lesson: 5.01 Quantitative Comparisons

A fair six-sided die is rolled. Find $P(\text{an odd number})$.

a. $\frac{2}{3}$

b. $\frac{1}{6}$

c. $\frac{1}{2}$ ✓

d. $\frac{1}{3}$

Odd outcomes $\{1, 3, 5\}$: $\frac{3}{6} = \frac{1}{2}$.

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41 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 6.01 Sequences & Patterns

A bag holds 4 red, 6 blue and 5 green marbles. Find $P(\text{not blue})$.

a. $\frac{3}{5}$ ✓

b. $\frac{9}{6}$

c. $\frac{2}{5}$

d. $\frac{6}{15}$

Not blue $= 4 + 5 = 9$ out of 15: $\frac{9}{15} = \frac{3}{5}$.

42 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 8.01 Word Problems & Reasoning

Two fair coins are tossed. Find $P(\text{at least one tail})$.

a. $\frac{3}{4}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

$P(\text{no tails}) = \frac{1}{4}$, so $P(\text{at least one}) = 1 - \frac{1}{4} = \frac{3}{4}$.

43 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 1.01 Arithmetic & Number Sense

A die is rolled twice. Find $P(\text{both rolls show an even number})$.

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

Each roll: $P(\text{even}) = \frac{1}{2}$; both: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

44 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A bag holds 5 white and 7 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both black})$.

a. $\frac{7}{22}$ ✓

b. $\frac{6}{11}$

c. $\frac{49}{144}$

d. $\frac{7}{12}$

$\frac{7}{12} \times \frac{6}{11} = \frac{42}{132} = \frac{7}{22}$.

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45 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 3.01 Algebra Essentials

A card is drawn, replaced, then a second card is drawn from a standard deck. Find $P(\text{both hearts})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{1}{16}$ ✓

$\left(\frac{13}{52}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$.

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